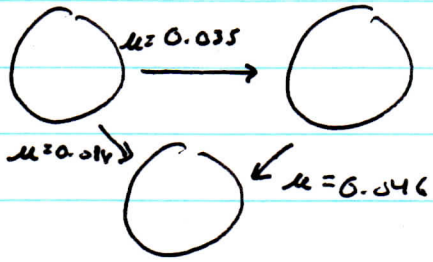
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Homework 2

1.



$$Z = \int_0^\infty (e^{-0.05t})(7500 e^{-(0.035+0.011)t})\,dt + \int_0^\infty (e^{-0.05t})(5000 e^{-(0.035+0.011)t})(0.011)\,dt + \cdots$$

$${}_t p_x^{01} = \int_0^t e^{-(0.035+0.011)s}(0.035)\,e^{-0.046(t-s)}\,ds$$

$$= 0.035 e^{-0.046t}\int_0^t e^{(0.046-0.053)s}\,ds = 0.035 e^{-0.046t}\int_0^t e^{-0.007s}\,ds$$

$$= \left(\frac{0.035}{0.007}\right)\left(e^{-0.046t}\right)\left(1-e^{-0.007t}\right) = 5\left(e^{-0.046t} - e^{-0.053t}\right)$$

$$\Rightarrow Z = \int_0^\infty (e^{-0.05t})(7500 e^{-0.053t})\,dt + \int_0^\infty (e^{-0.05t})(5000 e^{-0.053t})(0.011)\,dt + \int_0^\infty (e^{-0.05t})(5000)(5)(e^{-0.046t} - e^{-0.053t})(0.046)\,dt$$

$$= \frac{7500}{0.053} e^{-0.053t}\Big|_\infty^0 + \frac{900}{0.053} e^{-0.053t}\Big|_\infty^0 + \frac{11500}{0.051} e^{-0.051t}\Big|_\infty^0 - \frac{11500}{0.053} e^{-0.053t}\Big|_\infty^0$$

$$= \left(\frac{7500 + 900 - 11500}{0.053}\right) + \left(\frac{11500}{0.051}\right) = \boxed{172041.92 = Z}$$

2. policy: 12500 in 0

95000 @ 2

40000 if 1 @ $t=10$.

250 + 0.1G at issue

$$G = (0.1G + 250) + 12500(0.0395) + 40000\left(\frac{0.60875}{1.05^{10}}\right) + 95000(0.56810)$$

$$\Rightarrow G = \frac{1}{0.9}(250 + 493.75 + 14949.52 + 53969.5) = \boxed{77403.08 = G}$$

3. a. $${}_{10.25}p_x^{(t)} = e^{-\int_0^{10.25}\left(\frac{1+3}{45-t}\right)dt}$$

$u = 45-t$, $du = -dt$

$$\Rightarrow -\int_0^{10.25}\left(\frac{4}{45-t}\right)dt = 4\int_0^{10.25}\frac{1}{u}du = 4\ln(45-t)\Big|_{t=0}^{t=10.25}$$

$$= 4\ln\left(\frac{34.75}{45}\right) = \ln\left(\left(\frac{34.75}{45}\right)^4\right)$$

$$\Rightarrow {}_{10.25}p_x^{(t)} = e^{\ln((\cdot))} = \left(\frac{34.75}{45}\right)^4 = \boxed{0.3556 = {}_{10.25}p_x^{(t)}}$$

b. ~~${}_{2.5|4.7}q_x^{(2)} = {}_{7.2}q_x^{(2)} / {}_{4.7}q_x^{(2)}$~~ $= e^{-\int_{4.7}^{7.2}\frac{3}{45-t}dt}$

$$-\int_{4.7}^{7.2}\frac{3}{45-t}dt = 3\ln(45-t)\Big|_{t=4.7}^{t=7.2}$$

3.6. Wants ${}_{2.5|4.7}q_x^{(2)}$. $\mu_{x+t}^{(1)} = \frac{1}{45-t}$, $\mu_{x+t}^{(2)} = \frac{3}{45-t}$

~~${}_tp_x^{(\tau)} \cdot \mu_{x+t}$~~ ${}_tp_x^{(\tau)} = e^{-\int_0^t \frac{4}{45-s}\,ds} = e^{-4\ln(45)-}$

$u = 45-s$, $du = -ds$

$\Rightarrow {}_tp_x^{(\tau)} = e^{4\int_0^t u\,du} = e^{4(\ln(45-t)-\ln(45))}$

$= e^{\ln\left(\left(\frac{45-t}{45}\right)^4\right)} = \left(\frac{45-t}{45}\right)^4$

$\Rightarrow f_{T,J}(2) = \left(\frac{45-t}{45}\right)^4\left(\frac{3}{45-t}\right) = \left(\frac{1}{15}\right)\left(\frac{45-t}{45}\right)^3$

${}_{2.5|4.7}q_x^{(2)} = \int_{4.7}^{4.7+2.5} f_{T,J}(2)\,dt = \left(\frac{1}{15\times 45^3}\right)\int_{4.7}^{7.2}(45-t)^3\,dt$

~~$\left(\frac{1}{15\times 45^3}\right)$~~ $u = 45-t$, $du = -dt$

$= -\left(\frac{1}{15\times 45^3}\right)\int_{4.7}^{7.2} u^3\,du = -\left(\frac{1}{60\times 45^3}\right)\left[u^4\right]_{4.7}^{7.2}$

$= \left(\frac{1}{60\times 45^3}\right)\left((45-4.7)^4-(45-7.2)^4\right)$

$\boxed{{}_{2.5|4.7}q_x^{(1)} = 0.109}$

4. a. $\boxed{{}_5P_{45}^{(\tau)} = \frac{\ell_{50}}{\ell_{45}} = \frac{117145.49}{130718.70} = 0.8962}$

b. ~~${}_4q_{51}^{(2)} = \frac{d_{55}}{\ell_{51}} = \frac{103.50}{\text{[illegible]}}$~~

$\boxed{{}_4q_{51}^{(2)} = \frac{d_{51}+d_{52}+d_{53}+d_{54}}{\ell_{51}} = 0.00383}$

c. $\mathbb{P}(J(62)=3) = \frac{r_{62}+r_{63}+r_{64}+r_{65}}{\ell_{62}} = \frac{5017.51+4515.21+4061.64+3848.26}{52855.6}$

$= \boxed{0.9853 = \mathbb{P}(J(62)=3)}$

5. $d_{15k}^{(\tau)} \neq 0 \in \{0,\dots,38\} \Rightarrow 39$ nonzero failure classes

a. $\Rightarrow 45\times 39 + g\times 39 = 2223 \Rightarrow \boxed{g=12.}$

b. ~~${}_5|q_{50}^{(1)} = \frac{12\times 5}{2223-5(12+45)} = \frac{60}{\text{[illegible]}} = 0.310 = {}_5|q_{50}^{(1)}$~~

a

${}_5|q_{50}^{(1)} = ({}_8P_{50}^{(\tau)})({}_3q_{58}^{(1)}) = \left(\frac{2223-(5+8)(12+45)}{2223-5(12+45)}\right)\left(\frac{5q}{2223-(5+8)(12+45)}\right)$

$= \frac{5\times 12}{2223-5(57)} = \boxed{0.310 = {}_5|q_{50}^{(1)}}$

c. $\mathbb{P}(K(90)=4 \cap J(90)=1) = \mathbb{P}(\text{(90) lives 4 years and leaves by cause 1})$

$= \left(\frac{2223-15\times(12+45)}{2223-15\times(12+45)}\right)\left(\frac{12}{2223-15\times(12+45)}\right) = \frac{12}{2223-15\times 57} = 0.0088$

$\boxed{\mathbb{P}(K(50)=4 \cap J(90)=1) = 0.0088}$